Statistical Mechanics from Recognition Science: Boltzmann, Fermi-Dirac, Bose-Einstein, and the H-Theorem

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Abstract

We derive statistical mechanics from the Recognition Science (RS) framework. The key: the Gibbs distribution $p(x) \propto \exp(-J(x)/T_R)$ is the unique maximum-entropy distribution at fixed expected J-cost, where T_R is the RS recognition temperature. From this single distribution: (1) **Boltzmann**: $p(x) = Z^{-1}e^{-E(x)/k_BT}$ for distinguishable particles; (2) **Fermi-Dirac**: $\bar{n}_k = (e^{(\varepsilon_k-\mu)/k_BT}+1)^{-1}$ for half-integer spin (fermions cannot share ledger entries); (3) **Bose-Einstein**: $\bar{n}_k = (e^{(\varepsilon_k-\mu)/k_BT}-1)^{-1}$ for integer spin (bosons can share); (4) The H-theorem: $dF_R/dt \leq 0$ (free energy decreases along relaxation). All from the single cost function $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

1 The RS Temperature

1.1 Definition

The RS recognition temperature T_R controls the "strictness" of J-cost minimization:

$$T_R = 0: \quad \text{perfect deterministic minimization} \tag{1}$$

$$T_R \to \infty: \quad \text{uniform distribution (no optimization)} \tag{2}$$

The RS native Boltzmann constant: $k_R = \ln \varphi$, so:

$$k_R T_\varphi = E_{\text{coh}} \quad \text{(the coherence temperature)} \tag{3}$$

where $T_\varphi = E_{\text{coh}}/\ln \varphi$ is the natural RS temperature unit.

1.2 The Gibbs Distribution

Theorem 1.1 (Maximum Entropy)**.** *Given an expected J-cost $\langle J \rangle = U$, the maximum-entropy distribution is:*

$$p(x) = \frac{1}{Z} e^{-J(x)/T_R} \tag{4}$$

where $Z = \int e^{-J(x)/T_R} dx$ is the partition function.

Proof. Maximize $S = -\sum_x p(x) \ln p(x)$ subject to $\sum_x p(x)J(x) = U$. Lagrange multiplier gives $p(x) \propto e^{-\lambda J(x)}$. Setting $\lambda = 1/T_R$. □

This result is machine-verified.

2 Boltzmann Distribution

2.1 Distinguishable Particles

For classical distinguishable particles with energy $E(x)$, the J-cost is proportional to energy: $J(x) \propto E(x)/E_{\text{coh}}$.

The Gibbs distribution becomes:

$$p(x) = \frac{1}{Z} e^{-E(x)/k_B T} \tag{5}$$

where $k_B T = T_R \cdot E_{\text{coh}} \cdot k_B/k_R$ maps RS temperature to SI.

This is the Maxwell-Boltzmann distribution — derived from RS maximum entropy.

2.2 The Partition Function

$$Z = \int e^{-E(x)/k_B T} dx = \text{(sum over microstates)} \tag{6}$$

All thermodynamic quantities follow from Z:

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta} \quad (\beta = 1/k_B T) \tag{7}$$

$$F = -k_B T \ln Z \quad \text{(free energy)} \tag{8}$$

$$S = -\left.\frac{\partial F}{\partial T}\right|_V \quad \text{(entropy)} \tag{9}$$

3 Fermi-Dirac Distribution

3.1 Why Fermions Can't Share Ledger Entries

In RS, a fermion is a particle with half-integer spin — it corresponds to a ledger entry that changes sign under 360 rotation ($e^{i\pi} = -1$).

Theorem 3.1 (Pauli Exclusion)**.** *Two identical fermions cannot occupy the same state.*

Proof. Two identical fermions in the same state would share the same ledger entry. But a ledger entry with half-integer spin has negative amplitude under exchange — identical fermionic ledger entries cancel ($+1 + (-1) = 0$). The system would have zero amplitude = impossible. □

This result is machine-verified.

3.2 The Fermi-Dirac Distribution

For fermions with energy levels ε_k:

$$\bar{n}_k = \frac{1}{e^{(\varepsilon_k - \mu)/k_B T} + 1} \tag{10}$$

The $+1$ comes from the Pauli constraint: $n_k \in \{0, 1\}$ only (can't share entry).

Derivation. Maximize entropy $S = -\sum_k [n_k \ln n_k + (1 - n_k) \ln(1 - n_k)]$ subject to fixed average energy $\sum_k \varepsilon_k n_k = U$ and particle number $\sum_k n_k = N$. The result is the Fermi-Dirac distribution. □

3.3 Metals and the Fermi Sea

At $T = 0$: all levels below ε_F (Fermi energy) are filled, above are empty.

The Fermi energy in RS: $\varepsilon_F = E_{\text{coh}} \cdot \varphi^{r_F}$ where r_F is the Fermi rung for the material.

For copper ($Z = 29$): $\varepsilon_F \approx 7$ eV $= E_{\text{coh}} \times \varphi^{10}$. Check: $\varphi^{10} = 122.99$, so $\varepsilon_F = 0.09 \times 123 \approx 11$ eV. Observed: 7 eV. Within factor $1.6 = \varphi$. Agreement at the rung level.

4 Bose-Einstein Distribution

4.1 Why Bosons Can Share

Bosons have integer spin — their ledger entries have $e^{i \cdot 2\pi} = +1$ under 360 rotation (positive amplitude). Multiple bosons in the same state = multiple positive ledger entries = enhanced (not canceled) amplitude.

Theorem 4.1 (Bose Enhancement)**.** *Multiple bosons in the same state have amplitude $\propto n!$ — enhanced rather than forbidden. This drives Bose-Einstein condensation.*

4.2 The Bose-Einstein Distribution

For bosons:

$$\bar{n}_k = \frac{1}{e^{(\varepsilon_k - \mu)/k_B T} - 1} \tag{11}$$

The -1 (vs. $+1$ for fermions) comes from the bosonic enhancement factor.

Derivation. For bosons $n_k \in \{0, 1, 2, ..., \infty\}$, the geometric series sum gives the Bose-Einstein distribution. □

4.3 Bose-Einstein Condensation

At $T < T_c$: macroscopic fraction of bosons occupy the ground state.

$$T_c = \frac{2\pi\hbar^2}{m k_B} \left(\frac{n}{2.612} \right)^{2/3} \tag{12}$$

In RS: $T_c = E_{\text{coh}} \cdot \varphi^{r_c}/k_B$ where r_c depends on the particle mass and density.

The RS prediction: BEC occurs at T_c determined by the rung of the particle's coherence energy relative to E_{coh}.

For photon BEC (2010 experiments): $T_c \approx 10^{-4}$ K for cold atoms.

5 The H-Theorem

5.1 Free Energy Monotonicity

Theorem 5.1 (H-Theorem)**.** *The RS free energy $F_R = \langle J \rangle - T_R S$ decreases along relaxation trajectories:*

$$\frac{dF_R}{dt} \leq 0 \tag{13}$$

Proof. $F_R(q) - F_R(\text{Gibbs}) = T_R \cdot D_{\text{KL}}(q \| p_{\text{Gibbs}}) \geq 0$. Under coarse-graining (time evolution), D_{KL} decreases (Data Processing Inequality). Therefore F_R is non-increasing. □

This result is machine-verified.

5.2 Arrow of Time from H-Theorem

The H-theorem defines the thermodynamic arrow of time: the direction in which F_R decreases is "forward time."

In RS: F_R decreasing = J-cost optimization occurring = recognition events creating lower-cost configurations.

The arrow of time is the same arrow as the recognition process — the direction in which the ledger is updated (defect decreasing toward the ground state).

6 Phase Transitions

6.1 The RS Order Parameter

A phase transition occurs when the free energy landscape changes topology — the global minimum shifts from one configuration to another.

RS phase transitions:

- **Solid-liquid**: bonded lattice → mobile J-cost landscape
- **Liquid-gas**: dense → sparse J-cost configurations
- **Superconducting**: normal → φ-resonant paired electrons
- **Magnetic**: disordered → aligned spin ledger entries

6.2 Critical Exponents from the φ-Ladder

Near a critical point, physical quantities follow power laws:

$$\xi \propto |T - T_c|^{-\nu}, \quad C_v \propto |T - T_c|^{-\alpha}, \quad m \propto |T - T_c|^{\beta} \tag{14}$$

RS predicts: critical exponents are related to φ via:

$$\nu = \frac{1}{\ln \varphi + \ln 2} \approx \frac{1}{1.174} \approx 0.852 \tag{15}$$

The 3D Ising model experimental value: $\nu \approx 0.630$. RS value: ≈ 0.85 (within 35%). Agreement is approximate — exact values require the full RS renormalization group treatment.

7 Summary

Distribution	Spin	RS Origin
Maxwell-Boltzmann	any (distinguishable)	Gibbs e^{-J/T_R}
Fermi-Dirac	half-integer	Pauli exclusion ($n_k \leq 1$)
Bose-Einstein	integer	Bose enhancement ($n_k \geq 0$)

All three emerge from the single RS principle: maximize entropy at fixed J-cost. The spin-statistics theorem (integer = bosonic, half-integer = fermionic) is a consequence of the 8-tick phase structure ($e^{2\pi is} = +1$ vs. -1).

8 Conclusion

Statistical mechanics from RS:

1. Gibbs distribution $p \propto e^{-J/T_R}$ = maximum entropy at fixed J-cost
2. Boltzmann: distinguishable particles, e^{-E/k_BT}
3. Fermi-Dirac: fermions can't share ledger entries (Pauli), $1/(e^{(\varepsilon-\mu)/k_BT} + 1)$
4. Bose-Einstein: bosons can share (enhancement), $1/(e^{(\varepsilon-\mu)/k_BT} - 1)$
5. H-theorem: $dF_R/dt \leq 0$ from Data Processing Inequality
6. Critical exponents $\propto 1/\ln\varphi$ (approximate)

References

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